Christophe de Dinechin

Software as a Passion

LANGUAGES · SYSTEMS · TOOLS · GAMES · PHYSICS · COMMUNICATION SKILLS



ABOUT

Pushing the boundaries by writing highly innovative and differentiating software to make as much of a difference as possible.

French software developer — games, languages, operating systems

Amateur physics researcher

Tinkerer in strange forms of art

KEY SKILLS

- Languages & Compilers
- Systems & Virtualization
- Tools & Apps
- Games & 3D
- Physics
- Communication Skills

INFO

★ Tinkerer in strange forms of art

✉ christophe@dinechin.org

⌘ github.com/c3d

◉ c3d.github.io

TOOLS & TECHNOLOGIES

SoC · Firmware · System · Validation · Performance · Pre-silicon · C++ · C · RPL · Lisp · XL · Tao3D · ELFE · Linux · KVM · macOS · Windows · SPICE · Emacs · OpenShift · Make

Interactive constellation

https://c3d.github.io/cv/

LANGUAGES & COMPILERS

HP aC++ Compiler 1998-2000

- Wrote the first native Itanium C++ compiler before silicon; ran it on Ski (HP's functional Itanium simulator); HP's plan was a PA-RISC cross-compiler via Aries

DB48X automated test suite 2022-Present

- Built-in simulator suite in src/tests.cc / src/tests.h (3412 named step checks; F12 / -Tall)

DB48X - Modern RPL Calculator 2022-Present

- Rebuild and improve the HP48-family user experience

Native Itanium C++ compiler bring-up 1998-2000

- Wrote the first native Itanium C++ compiler before silicon existed; ran it on Ski, HP's functional Itanium simulator (a second simulator also modelled performance details)

SYSTEMS & VIRTUALIZATION

HPVM - Enterprise Virtualization 2000-2010

- Early HPVM boot entirely on Ski (pre-silicon); later A-step silicon bring-up

HPVM pre-silicon and A-step bring-up 2000-2010

- Early HPVM (HP Integrity VM) boot was done entirely on Ski, HP's functional Itanium simulator — physical hardware took a while to arrive

HP ECUTEST Real-Time System 1994-1998

- Designed HP ECUTEST for engine control unit testing and system validation

SPICE Remote Streaming 2017-Present

- Part of the Spice team at Red Hat

TOOLS & APPS

Make-It-Quick

- Short readable makefiles without automake/cmake complexity

Recorder

- Continuous recording with printf-like statements for performance and debug traces

Emacs on Aqua

- Adapted an old NeXT port to Aqua when only Terminal Emacs shipped

GAMES & 3D

Alpha Waves / Continuum 1989-1990

- Started in 1988, published by Infogrames in 1990

HP-48 Games 1989-1993

- PacMan — first use of HP48 hardware scroll (per Paul Courbis)

Taodyne 2010-2016

- Created Tao3D to present in real-time 3D (big data, movies, glasses-free 3D)

PHYSICS

A Theory of Incomplete Measurements

- A Theory of Incomplete Measurements — unifying GR and quantum mechanics

Réunifions la physique

- Gentle French introduction to incomplete-measurement physics

COMMUNICATION SKILLS

3DIVART

- Adapt ray-tracing / ray-marching shaders for auto-stereoscopic screens via Tao3D

Informagie

- Started as a student around the same time as Alpha Waves